

## Chapter 11 Homework

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### Part A multiple choice (13 points)

1-5 EBBEC

6-10 DDCAA

11-13 BAE

### Part B short question (17 points)

14 (2 points)

(a) (1 point) The cis isomer has stronger dipole-dipole forces.

(b) (1 point) The cis isomer boils at 60.3 °C and the trans isomer boils at 47.5 °C.

15 (3 points)

$$\Delta H_1 = mc_{s1}\Delta T_1 = 42.0 \times 0.970 \times (155 - 114) \div 1000 = 1.67\text{kJ}$$

$$\Delta H_2 = n \cdot \Delta H_{\text{fus}} = \frac{42.0}{46.0} \times 5.02 = 4.58\text{kJ}$$

$$\Delta H_2 = mc_{s2}\Delta T_2 = 42.0 \times 2.30 \times (78 + 114) \div 1000 = 18.55\text{kJ}$$

$$\Delta H_4 = n \cdot \Delta H_{\text{vap}} = \frac{42.0}{46.0} \times 38.56 = 35.21\text{kJ}$$

$$\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3 + \Delta H_4 = 1.67 + 4.58 + 18.55 + 35.21 = 60.0\text{kJ}$$

16 (4 points)

CF<sub>4</sub>, CClF<sub>3</sub>, CCl<sub>2</sub>F<sub>2</sub>, CCl<sub>3</sub>F

17 (4 points)

(a)

step 1: H<sub>2</sub>O(*g*) will **condense** to H<sub>2</sub>O(*s*);

step 2: H<sub>2</sub>O(*s*) will **melt** to form H<sub>2</sub>O(*l*).

(H<sub>2</sub>O(*g*) will condense to H<sub>2</sub>O(*s*) at approximately 4 torr; at a higher pressure, perhaps 5 atm or so, H<sub>2</sub>O(*s*) will melt to form H<sub>2</sub>O(*l*).)

(b)

step 1: H<sub>2</sub>O(*g*) will **condense** to H<sub>2</sub>O(*l*);

step 2: H<sub>2</sub>O(*l*) will **freeze** to form H<sub>2</sub>O(*s*).

(At 100 °C and 0.50 atm, water is in the vapor phase. As it cools, water vapor condenses to the liquid at approximately 82 °C, the temperature where the vapor pressure of liquid water is 0.50 atm. Further cooling results in freezing at approximately 0 °C. The freezing point of water increases with decreasing pressure, so at 0.50 atm the freezing temperature is very slightly above 0 °C.)

18 (4 points)

diethyl ether > acetone > ethanol > water